

Hints & Solutions

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Q1 (4) - Exponent of prime

Clearly, $105! + 2$ is divisible by 2, as

$$105! + 2 = (1 \cdot 2 \cdot 3 \cdots 105) + 2 = 2((1 \cdot 3 \cdot 4 \cdots 105) + 1)$$

Similarly, the numbers $105! + 3, 105! + 4, \dots, 105! + 105$ are divisible by 3, 4, 5, ..., 105, respectively. So,

none of the given numbers are prime, as prime number has factors only 1 and itself.

Q2 (2) - Exponent of prime

Since, perfect square of every natural number ends with

$\{0, 1, 4, 5, 6, 9\}$, but square of any natural number cannot have unit's digit $\{2, 3, 7, 8\}$... (i)

Also, $1! + 2! + 3! + 4! + 5! + \dots + x!$, when $x \geq 4$ units place ends with 3

Thus, $x \in \{1, 2, 3\}$

Now, if $x = 1 \Rightarrow 1! = 1$ [perfect square]

if $x = 2 \Rightarrow 1! + 2! = 3$ [cannot be perfect square]

if $x = 3 \Rightarrow 1! + 2! + 3! = 9$ [perfect square]

Thus, x can take value 1 and 3.

$\therefore$ Integral solution for x is two.

Q3 (1) - Restricted Arrangements

The given word 'BANANA' contains 6 letters out of which A occur thrice. N occur twice and rest are distinct.

$$\therefore \text{Total number of arrangements} = \frac{6!}{3!2!} = 60$$

When two N's are appear adjacently, treated as one letter.

We have to arrange 5 letters out of which A occur thrice and rest are distinct

$$\therefore \text{Required number of arrangement} = \frac{5!}{3!} = 20$$

$\therefore$ The total number of word two N's do not appear adjacently =

(Total number of words) - (Number of words in which two N's appear adjacently)

$$= 60 - 20 = 40$$

Hence, option (1) is correct.

Hints & Solutions

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Q4 (3) - Restricted Arrangements

There are nine-digits numbers 223355888 out of which 5 are even number and 4 are odd numbers.

Let us mark out the position to filled up as follows

(1)(2)(3)(4)(5)(6)(7)(8)(9)

Now, the 4 odd number can be placed at any of the four place, marked (2,4,6,8)

So, the number of ways of arranging the digit (3,3,5,5) out of which 3 and 5 occur twice

$$= \frac{4!}{2!2!} = 6$$

Also, the 5 even numbers (2,2,8,8,8) out of which 2 occurs twice and 8 occurs thrice may be arranged

$$= \frac{5!}{2!3!} = 10$$

$\therefore$ The required number of ways = $6 \times 10 = 60$ ways

Q5 (1) - Restricted Arrangements

There are 18 beads out of which 8 being blue, 5 red, 4 green and 1 yellow. In this arrangement anti-clockwise or clockwise are not different.

$$\therefore \text{Total number necklace formed} = \frac{(18-1)!}{2!8!5!4!} = \frac{17!}{2 \cdot 8! \cdot 5! \cdot 4!}$$

Hence, option (1) is correct.

Q6 (3) - Restricted Arrangements

There are 11 letters in word 'MISSISSIPPI' out of which I occurs 4 times, S occurs 4 times and P occurs twice and rest are different.

Consider four S as one letter. We have to arrange 8 letters out of which I occurs 4 times and P occur twice and rest are distinct.

$$\therefore \text{Total number of arrangement} = \frac{8!}{4!2!}$$

Hence, option (3) is correct.

Q7 (2) - Restricted Arrangements

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Since, at most one ring is to be worn in any four fingers.

$\therefore$ Total number of arrangement of worn in four fingers 4P_3

$$= \frac{4!}{(4-3)!} = 4! = 24 \text{ ways}$$

Hence, option (2) is correct.

Q8 (3) - Restricted Arrangements

Since, atleast one ring is to be worn in each finger,

So, we first select four rings out of 5 given rings and then they are arranged in fingers. This can be done in

$${}^5P_4 = 5! = 120 \text{ ways.}$$

Now, one ring is left which can be worn in anyone of the four finger in 4 ways.

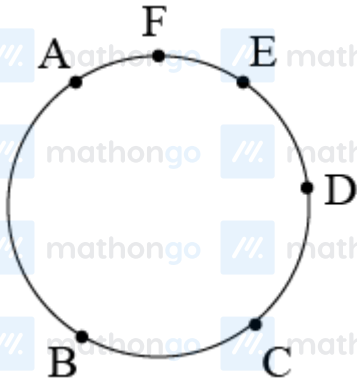
$\therefore$ Required number of ways $= 120 \times 4 = 480$ ways.

Hence, option (3) is correct.

Q9 (2) - Geometric Applications

Number of lines drawn through 6 points lying on a circle

$$= {}^6C_2 = \frac{6 \times 5}{2 \times 1} = 15$$

**Q10 (2) - Geometric Applications**

Here, total number of points are $(m + n + k)$ which must give ${}^{m+n+k}C_3$ number of triangles, but m points on l , taking 3 points at a time gives mC_3 combinations which produce no triangle. Similarly, nC_3 and kC_3 number of triangles can not be formed.

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Therefore, the required number of triangles are

$${}^{m+n+k}C_3 - {}^mC_3 - {}^nC_3 - {}^kC_3$$

Q11 (2) - Geometric Applications

Given, total number of points = 18 and number of collinear points = 5

We know that, one line has two ends points. Therefore, total number of lines = ${}^{18}C_2$

Since, 5 points are collinear, therefore total number of lines drawn from collinear points = ${}^5C_2 = 10$

Also, there is a line corresponding to the line of collinearity.

$$\begin{aligned}\therefore \text{Required number of lines} &= {}^{18}C_2 - {}^5C_2 + 1 \\ &= 153 - 10 + 1 = 144\end{aligned}$$

Q12 (2) - Geometric Applications

Since, two straight lines intersect at one point. Maximum number of points of intersections = 8C_2

$$= \frac{8 \times 7}{2 \times 1} = 28$$

Q13 (3) - Geometric Applications

Number of points required for the fixed circle is 3. So, first select any three points from the 10 points in ${}^{10}C_3$ ways.

In these ways, circle with four concyclic points is selected in 4C_3 ways. But it should be taken once.

$$\begin{aligned}\therefore \text{Total number of circles} &= ({}^{10}C_3 - {}^4C_3) + 1 \\ &= \left(\frac{10 \times 9 \times 8}{3 \times 2 \times 1} - 4 \right) + 1 \\ &= (120 - 4) + 1 = 117\end{aligned}$$

Q14 (2) - Geometric Applications

We know that, two circles intersect at two distinct points, two straight lines intersect at one point, one circle and one line intersect at two distinct points, therefore total number of points of intersection are as follows.

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Number of ways of selection	Point of intersection
Two straight lines : 4C_2	${}^4C_2 \times 1 = 6$
Two circles : 4C_2	${}^4C_2 \times 2 = 12$
One line and one circle : ${}^4C_1 \times {}^4C_1$	${}^4C_1 \times {}^4C_1 \times 2 = 32$

Thus, total number of point of intersection = $32 + 12 + 6 = 50$

Q15 (3) - Geometric Applications

Let number of sides of polygon = n

$$\therefore \frac{(n-2)180^\circ}{n} = 150$$

[$\because$ interior angle of n sided polygon is given by $\frac{(n-2)}{n}180^\circ$]

$$\Rightarrow (n-2)6 = 5n$$

$$\Rightarrow 6n - 12 = 5n$$

$$\Rightarrow n = 12$$

$$\begin{aligned}\therefore \text{Number of diagonals} &= {}^{12}C_2 - 12 \\ &= 6 \times 11 - 12 \\ &= 66 - 12 = 54\end{aligned}$$

Q16 (2) - Geometric Applications

Clearly, there are three horizontal lines and four vertical lines.

As, we know that a parallelogram can be formed by two vertical lines and two horizontal lines.

$$\begin{aligned}\therefore \text{Required number of parallelogram} &= {}^3C_2 \times {}^4C_2 \\ &= 3 \times \frac{4 \times 3}{2 \times 1} = 18\end{aligned}$$

Q17 (3) - Geometric Applications

Let the given box is of size 4×3 .

Now, on considering block of size 1×1 , we get $4 \times 3 = 12$ squares (number of ways of selecting a unit length horizontally $\times$ number of ways of selecting a unit length vertically). On considering block of size

2×2 , we get 3×2

$= 6$ squares (number of ways of selecting a length of unit 2 horizontally $\times$ number of ways of selecting a length of unit 2 vertically)

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On considering block of size 3×3 , we get $2 \times 1 = 2$ squares

$\therefore$ Total number of squares = $12 + 6 + 2 = 20$

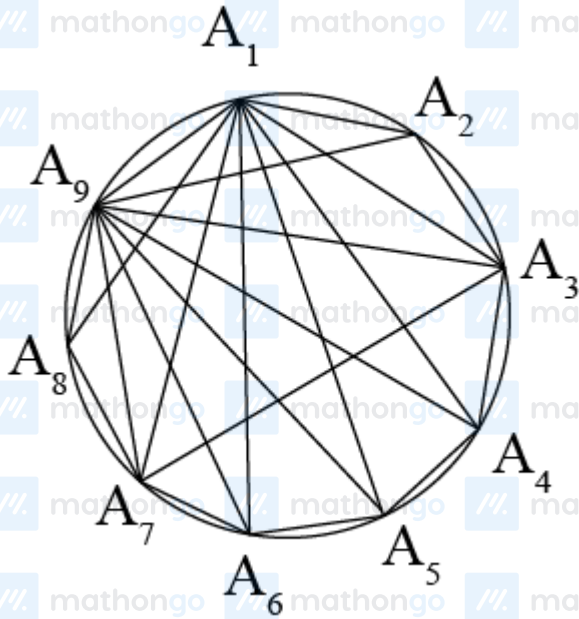
Q18 (4) - Geometric Applications

Total number of points are 18 which must give $^{18}C_3$ number of triangles, but on each side of triangles 7 points are lying. Now, on taking 3 points at a time from any one side gives 7C_3 combinations which produce no triangle.

$$\begin{aligned} \text{Required number of triangle} &= ^{18}C_3 - ^7C_3 - ^7C_3 - ^7C_3 \\ &= \frac{18 \times 17 \times 16}{3 \times 2 \times 1} - 3 \times \frac{7 \times 6 \times 5}{3 \times 2 \times 1} \\ &= 816 - 105 = 711 \end{aligned}$$

Q19 (1) - Geometric Applications

The number of points of intersection of straight lines inside the polygon = number of intersection points of diagonals inside the polygon = $^9C_4 = 126$



Q20 (2) - Division in groups

Here, 9 students are to be divided into 3 equal groups and the order of the groups are important. So, required

$$\text{number of ways} = \frac{9!}{(3!)^3 \cdot 3!} \times 3! = \frac{9!}{(3!)^3}$$

Hence, option (2) is correct.

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Q21 (2) - Division in groups

First group can be given 8 different things out of 15 in ${}^{15}C_8$ ways second group can be given 4 different thing out remaining 7 in 7C_4 and 3 different things out of 3 remaining 3 things in 3C_3 ways.

Hence, the required number of ways (here, order of groups are not important)

$$= {}^{15}C_8 \times {}^7C_4 \times {}^3C_3 = \frac{15!}{8!7!} \times \frac{7!}{4!3!} \times \frac{3!}{3!} = \frac{15!}{8!4!3!}$$

Hence, option (2) is correct.

Q22 (3) - Division in groups

Here, there are 12 different balls and three different boxes the first box contain exactly 5 balls out of 12 ball is ${}^{12}C_5$ ways and the remaining 7 balls have two boxes for each of 7 balls i.e.

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^7 \text{ ways}$$

$$\therefore \text{Required number of ways} = {}^{12}C_5 \times 2^7 \text{ ways}$$

Hence, option (3) is correct.

Q23 (2) - Division in groups

There are 50 identical things distributed among 8 persons. In which 3 of them get 8 things each, two of them get 7 things each and remaining 3 get 4 thing each

$$8 + 8 + 8 + 7 + 7 + 4 + 4 + 4 = 50$$

$$3(8) + 2(7) + 3(4) = 50$$

Now, number of ways to distribute $(3 + 2 + 3)$ items among 8 persons in the groups 3, 2, and 3 items

$$= \frac{(3+2+3)!}{3!2!3!} = \frac{8!}{(3!)^2(2!)}$$

Hence, option (2) is correct.

Q24 (3) - Division in groups

We have, 'MISSISSIPPI' words contain I occurs 4 times, S occurs 4 times, P occurs twice and M are one time,

$$M = 1, I = 4, S = 4, P = 2$$

Zero or more 'I' can be selected out of 4I's $= (4 + 1) = 5$ ways Zero or more 'S' can be selected out of

$$4 S's = (4 + 1) = 5 \text{ ways Zero or more 'P' can be selected out of } 2P's = (2 + 1) = 3 \text{ ways}$$

Zero or more 'M' can be selected out of $M's = (1 + 1) = 2$ ways

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So, total number of selection = $5 \times 5 \times 3 \times 2 = 150$ ways

But in one of these selection number of each type of letter is zero. Hence, this selection must be excluded.

$\therefore$ Total number of selection = $150 - 1 = 149$ ways

Hence, option (3) is correct.

Q25 (2) - Division in groups

Here, 200 different things are to be divided in 100 equal groups.

$$\begin{aligned}\therefore \text{The total numbers of ways} &= \frac{200!}{(2!)^{100} \times (100)!} \\ &= \frac{200!}{(100)!2^{100}}\end{aligned}$$

Hence, option (2) is correct.

Q26 (2) - Division in groups

Here, seven different coins are to be divided amongst three person in which no two persons receive the same number of coin but each receives at least one coin.

The possible number of coin distributed is (1,2,4).

$$\therefore \text{Total number of ways} = \frac{7!}{1!2!4!} \times 3! = 630$$

Hence, option (2) is correct.

Q27 (1) - Number of Divisors and Sum of divisors

$$\text{We have, } n = 21600 = 2^5 \times 3^3 \times 5^2$$

A divisors of 21600 will be a multiple of 10 but not 15. We must have at least one 2 out of 5, 2's and at least one 5 out of 2 but 3 is not taken. Hence, the number of divisor which are multiple of 10 but not 15 is

$$(5)(2) = 10 \text{ ways. Hence, option (a) is correct.}$$

Q28 (4) - Number of Divisors and Sum of divisors

$$\text{We have, given } 2^5 \cdot 3^7 \cdot 5^3 \cdot 7^2$$

The sum of the divisor is

Hints & Solutions

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$$\left(\frac{2^6-1}{2-1}\right) \left(\frac{3^8-1}{3-1}\right) \left(\frac{5^4-1}{5-1}\right) \left(\frac{7^3-1}{7-1}\right)$$

$$= \frac{(2^6-1)(3^8-1)(5^4-1)(7^3-1)}{2 \times 4 \times 6}$$

Hence, option (4) is correct.

Q29 (3) - Number of Divisors and Sum of divisors

We have, $N = 1000$

$$\Rightarrow N = 2^3 \times 5^3$$

The even divisor of 1000

2, 4, 8, 10, 20, 40, 50, 100, 200, 250, 500, 1000

the product of even divisor is

$$2 \times 4 \times 8 \times 10 \times 20 \times 40 \times 50 \times 100 \times 200 \times 250 \times 500 \times 1000$$

$$2^{10} \times 10^{14} \times 5^4$$

$$2^6 \times 2^4 \times 5^4 \times 10^{14} = 2^6 \times 10^{18} = 64 \times 10^{18}$$

Hence, option (3) is correct.

Q30 (2) - Number of Divisors and Sum of divisors

We have, given digits 1, 3, 5, 7, 9

Sum of number of one-digit = $1 + 3 + 5 + 7 + 9 = 25$ Two-digit numbers are formed taken in pair

(1, 3)(1, 5)(1, 7)(1, 9)(3, 5)(3, 7)(3, 9)(5, 7)(5, 9)(7, 9)

Sum of two-digit numbers are $25 \times 4(11) = 1100$ Three-digit numbers are formed (1, 3, 5)(1, 3, 7)(1, 3, 9)(1, 5, 7)(1, 5, 9)(1, 7, 9)

(3, 5, 7)(3, 5, 9)(3, 7, 9)(5, 7, 9)

Sum of three-digit numbers = $25 \times 6 \times 2 \times (111) = 33300$

Four-digits numbers are

(1, 3, 5, 7)(1, 3, 5, 9)(1, 5, 7, 9)(3, 5, 7, 9)(1, 3, 7, 9)

Sum of four-digit numbers are $25 \times 4 \times 6 \times (1111) = 666600$ Sum of all the numbers from the digit 1, 3, 5, 7, 9 which are small than 10000, if repetition of digit is not allowed are

$$25 + 1100 + 33300 + 666600 = 701025$$

$$= 25(28041)$$

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Q31 (1) - Dearrangement, Multinomial selection and arrangement

We know that, if there are n letters and n corresponding envelopes, then the number of ways in which letters can be placed in the envelopes, so that no letter is placed in the correct envelope is

$$n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{(-1)^n}{n!} \right]$$

Here, $n = 10$

$$\therefore \text{Required number of ways} = 10! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{1}{10!} \right]$$

Q32 (1) - Dearrangement, Multinomial selection and arrangement

Put $x = a - 1, y = b - 1, z = c - 1, t = d - 1$, then $a, b, c, d \geq 0$ and

$$(a - 1) + (b - 1) + (c - 1) + (d - 1) = 20$$

$$a + b + c + d = 24$$

which is equivalent to the number of ways of distributing 24 identical objects into 4 different boxes, if empty boxes are allowed which is $24 + 4 - 1 C_{4-1} = {}^{27}C_3$

Q33 (2) - Dearrangement, Multinomial selection and arrangement

Let $x = p - 5, y = q - 5$ and $z = r - 5$, then $p, q, r \geq 0$ and the

given equation reduces to $p + q + r = 15$

Now, we have to find non-negative integral solution of Eq. (i). The total number of such solution is

$${}^{15+3-1}C_{3-1} = {}^{17}C_2 = \frac{17 \times 16}{2} = 136$$

Q34 (1) - Dearrangement, Multinomial selection and arrangement

Let $x = 0$. Then, the equation reduces to $y + z = 33$.

Now, number non-negative integral solution of the equation $y + z = 33$ is ${}^{33+2-1}C_{2-1} = {}^{34}C_1$

Let $x = 1$. Then, the equation reduces to $y + z = 30$.

Now, number of non-negative integral solution of the equation $y + z = 30$ is ${}^{30+2-1}C_{2-1} = {}^{31}C_1$

Similarly, we have $x = 2, 3, 4, \dots, 11$ and the corresponding number of non-negative integral solution are

$${}^{28}C_1, {}^{25}C_1, \dots, {}^1C_1.$$

$$\therefore \text{Total number of integral solution} = {}^{34}C_1 + {}^{31}C_1 + \cdots + {}^1C_1$$

Hints & Solutions

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$$= 34 + 31 + 28 + \dots + 1$$

$$= \frac{12}{2} [1 + 34] \quad \left[\because \text{sum of } n \text{ terms of an AP} = \frac{n}{2} (a + 1) \right]$$

$$= 6(35) = 210$$

Q35 (4) - Dearrangement, Multinomial selection and arrangement

We have, $x_1 x_2 x_3 x_4 x_5 = 240$, where x_1, x_2, x_3, x_4, x_5 are natural numbers

Now, by using prime factorisation of 240, we can write the given equation as

$$x_1 x_2 x_3 x_4 x_5 = 2^4 \times 3 \times 5$$

Now, we can assign 3 and 5 to any of 5 unknowns in 5×5 ways. We can assign 2^4 to x_1, x_2, x_3, x_4, x_5 in

$$a + b + c + d + e = 4$$

$$\Rightarrow 4 + 5 - 1 C_{5-1} = {}^8 C_4, \text{ ways}$$

$$\therefore \text{Required number of ways} = {}^8 C_4 \times 25 = 1750$$

Q36 (4) - Dearrangement, Multinomial selection and arrangement

In 'MORADABAD' we have 6 different types of letters 3 A's, 2 Ds and rest four different. We have, to form words of 4 letters.

Now, four letters word can be formed in following ways

$$(i) \text{ All different number of ways } {}^6 P_4 = 6 \times 5 \times 4 \times 3 = 360$$

$$(ii) \text{ Two different two alike number of ways } {}^2 C_1 \times {}^5 C_2 \times \frac{4!}{2!} = 240$$

$$(iii) \text{ 3 alike 1 different number of ways } {}^1 C_1 \times {}^5 C_1 \times \frac{4!}{3!} = 20$$

$$(iv) \text{ 2 alike of one type and 2 alike of other type, number of ways } {}^2 C_2 \times \frac{4!}{2! \times 2!} = 6$$

$$\therefore \text{Total numbers of words} = 360 + 240 + 20 + 6 = 626$$

Q37 (1) - Dearrangement, Multinomial selection and arrangement

In 'COMBINATION' we have 8 different types of letters 2 Is, 2 O's, 2 N's and rest 5 different.

We have to form words by taking two same and two different letters and this can be done in ${}^3 C_1 \times {}^7 C_2 \times \frac{4!}{2!}$

$$\text{ways} = 3 \times 21 \times 12 = 756$$

Q38 (1) - Dearrangement, Multinomial selection and arrangement

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Given letters are I, (N, N, N), (D, D), (E, E, E), P, T

We have to select five letters from the given letters and this can be done in following ways.

(i) All the letters are distinct [different letters are I, N, D, E, P, T],

$$\text{number of ways} = {}^6C_5 = 6$$

(ii) 3 distinct and 2 alike, number of ways $= {}^3C_1 \times {}^5C_3 = 30$

$$\text{(iii) 2 distinct, 3 alike} = {}^2C_1 \times {}^5C_2 = 20$$

(iv) 2 alike, 2 alike, 1 distinct $= {}^3C_2 \times {}^4C_1 = 12$

$$\text{(v) 3 alike, 2 alike} = {}^2C_1 \times {}^2C_1 = 4$$

$\therefore$ Total number of ways of selecting 5 letters

$$= 6 + 30 + 20 + 12 + 4 = 72$$